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Lista de Exercícios

Baixe a ferramenta openssl para realizar as questões abaixo.

Submeta um relatório com capturas de tela de cada comando realizado.

Questão 1

Gere um par de chaves RSA de 2048 bits

private key:

```
➤ openssl genrsa -out private_key.pem 2048
```

```
-----BEGIN PRIVATE KEY-----
```

```
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBCwggSjAgEAAoIBAQCcYrYz15CzHyaaiz4u7ApKKGn7htxt+QT0xx/uTQUTEPeH3Ekhu3kHYVcbj0V+0XpkFX9bUtT2Bhr2NFJSIgQyY0Y05H/Pg2bDbdtITRFK39FZQ9YZOHfGa0MYC81E0E/OK0QsPTmAXF13kIrhgvxBTH3Bzx+ZVw2IXXVrEWTjuYH6H8vr31NbEV03zBrhDDuFD6QZdHYhoxZHC CNfH2Ld/GdNmEUmSJ4leOvXB8Qlbuw4cTnm3vubtYF/K37xHGzIu/lwCi01m/KNFJXutmd1EUP3z0zbbxGEWFrzdYiEXG4Q3glAzRnjObfsCmaBStRAd62zfYdcAu6puMYEoByArAgMBAAECggEAEKtDX/wOpDdmETjN/Xpqlle4ikXWuMI/Sxsq4n5fRFEG9oB8AQK/YfJSLHWjrcp1iXUtEXxs0nKRPlxvWmHrMVS7nETaoCy3QZa87oE982Xi oKSAZ89JcrN1fZqkFHv90dDrP14OtqavR7o/XWyuyGx9IgmX971vsbOttmnerIPC62UsvVeQmFj2x9af90uXBQJ2YImQc/bEPp5hJD7ZLq/w2VtiekgDGX1VE9eP8dCMHvEloS3+h2CM6bY4QLiIu6hEi+MCRrkdUHBjidAwJf6B4miqXZ12lbUPXKtX860M NlW8IMDyYuhZohpMsUyfOomce+XfAmbd7Y8dO/YXRQKBgQDQ2E5+XKAN239EAFiD3meETVumfOxZ1Gvq5+F9L2kaGz8A56NFJh+TdPhJcL1IZoXrotiWVlrvXkX37kx1h2Qk/HpdZbYU54UvCL9y7ND1SUXcFxPjQ6yaVVao4w1WjWxU367yaHpBCkDAuygbtXhc3eYBR/ZF6PzEBjyPP3j+jQKBgQC7JqrwnfA0uAKj69IFKH8snfWYP9qD5d0B JhzX+GBAJoImeQ38+Bd802oLm0g9DzZ19SePlnm+KlEe4qKeEzw7bAwR+aDfEA6IYJKjgLD3EAcighJPf/wBe6IfhSjR6117riwv/ACQAHd2H+AOGdLtLnKjjGlepb+XhHoDEsenlwKBgQCvCZxALEG5W45+4AtP+seoC24VS/6MC42nhFFpKqNcThmwGHSvMBSm8hJm+yap18PGiuYcmqPm0OyZr/Y34hH9Ld8hCG5jdugZU6uiFvnLm24B2RXC FKZFc+D+C8t3z3BMcm2enhJZEM7UuggTstUAILr1t+qhDXVml12Lohc2wQKBgBFT/zfQJ9F5yCTNTODVYxEKfx1F+EGGIvRcRwdoI+FMr3hIdRcbEXGpUYRjMgKu6ef6DvOohxUoQ4ZeJzF9R/JoiqWPa2dTmUTftTDR/UeNORJrdDFz4CK1KfTRV6/WjyUAOXGEFADhVERqJMQGdOn5jcpHaC1eNGTQ60+HU9095AoGASPAQehGMymUlXgXB4i2povNM/uc6VJoVTJxYW+2kVomR+eSdIIPLHrOs1Ek/ecThmSirhPElesI+8WkHgIiws4MaYX6uw5y/ODXDKWDTDuNpnhq/LK4aVK1KqKMcVKxERG10ErwoX6ucUA7ILX4Bap/6tD66e7XGIyDvYBvVWqc=
```

```
-----END PRIVATE KEY-----
```

public key:

```
➤ openssl rsa -in private_key.pem -pubout -out public_key.pem
```

```
-----BEGIN PUBLIC KEY-----
```

```
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAmK2M5eQsx8mmos+LuwKSihp+4bcbfkeZscf7k0FE3j3h9xJibt5B2FXG49Fftf6ZBV/W1LU9gR69jRSUiIEMmNGNOR/zxtmw23bSE0Rst/RWUPWGTh3xmtDGAvNRNBPzitELD05gFxdd5CK4YL8QUx9wc8fmVcNiF11axFk47mB+h/L699TWxFdN8wa4Qw7hQ+kGXR2IaMWRwgjXx9i3fxnTZhFJkieJXjrlwfEJW7sOHE55t77m7WBfyt+8RxsyLv5cAotNZvyjRSV7rZg5RFD989M228RhFha83WIhFxuEN4JQM0Z4zm37ApmgUrUQHets32HXALuqbjGBKAcgKwIDAQAB
```

```
-----END PUBLIC KEY-----
```

Questão 2

Exiba módulo e expoente da chave pública e da chave privada geradas.

public key:

```
> openssl rsa -pubin -in public_key.pem -text -noout
```

Public-Key: (2048 bit)

Modulus:

```
00:98:ad:8c:e5:e4:2c:c7:c9:a6:a2:cf:8b:bb:02:
92:8a:1a:7e:e1:b7:1b:7e:41:33:b1:c7:fb:93:41:
44:de:3d:e1:f7:12:48:6e:de:41:d8:55:c6:e3:d1:
5f:b4:5e:99:05:5f:d6:d4:b5:3d:81:1e:bd:8d:14:
94:88:81:0c:98:d1:8d:39:1f:f3:c6:d9:b0:db:76:
d2:13:44:52:b7:f4:56:50:f5:86:4e:1d:f1:9a:d0:
c6:02:f3:51:34:13:f3:8a:d1:0b:0f:4e:60:17:17:
5d:e4:22:b8:60:bf:10:53:1f:70:73:c7:e6:55:c3:
62:17:5d:5a:c4:59:38:ee:60:7e:87:f2:fa:f7:d4:
d6:c4:57:4d:f3:06:b8:43:0e:e1:43:e9:06:5d:1d:
88:68:c5:91:c2:08:d7:c7:d8:b7:7f:19:d3:66:11:
49:92:27:89:5e:3a:f5:c1:f1:09:5b:bb:0e:1c:4e:
79:b7:be:e6:ed:60:5f:ca:df:bc:47:1b:32:2e:fe:
5c:02:8b:4d:66:fc:a3:45:25:7b:ad:98:39:44:50:
fd:f3:d3:36:db:c4:61:16:16:bc:dd:62:21:17:1b:
84:37:82:50:33:46:78:ce:6d:fb:02:99:a0:52:b5:
10:1d:eb:6c:df:61:d7:00:bb:aa:6e:31:81:28:07:
20:2b
```

Exponent: 65537 (0x10001)

private key:

```
> openssl rsa -pubin -in private_key.pem -text -noout
```

Public-Key: (2048 bit)

Modulus:

```
00:98:ad:8c:e5:e4:2c:c7:c9:a6:a2:cf:8b:bb:02:
92:8a:1a:7e:e1:b7:1b:7e:41:33:b1:c7:fb:93:41:
44:de:3d:e1:f7:12:48:6e:de:41:d8:55:c6:e3:d1:
5f:b4:5e:99:05:5f:d6:d4:b5:3d:81:1e:bd:8d:14:
94:88:81:0c:98:d1:8d:39:1f:f3:c6:d9:b0:db:76:
d2:13:44:52:b7:f4:56:50:f5:86:4e:1d:f1:9a:d0:
c6:02:f3:51:34:13:f3:8a:d1:0b:0f:4e:60:17:17:
5d:e4:22:b8:60:bf:10:53:1f:70:73:c7:e6:55:c3:
62:17:5d:5a:c4:59:38:ee:60:7e:87:f2:fa:f7:d4:
d6:c4:57:4d:f3:06:b8:43:0e:e1:43:e9:06:5d:1d:
88:68:c5:91:c2:08:d7:c7:d8:b7:7f:19:d3:66:11:
49:92:27:89:5e:3a:f5:c1:f1:09:5b:bb:0e:1c:4e:
79:b7:be:e6:ed:60:5f:ca:df:bc:47:1b:32:2e:fe:
5c:02:8b:4d:66:fc:a3:45:25:7b:ad:98:39:44:50:
fd:f3:d3:36:db:c4:61:16:16:bc:dd:62:21:17:1b:
84:37:82:50:33:46:78:ce:6d:fb:02:99:a0:52:b5:
10:1d:eb:6c:df:61:d7:00:bb:aa:6e:31:81:28:07:
20:2b
```

Exponent: 65537 (0x10001)

Questão 3

Gere um certificado autoassinado válido por 365 dias usando a chave pública que você gerou anteriormente.

Utilize o formato .PEM.

Ao tentar com chave pública:

```
> openssl req -x509 \
-key public_key.pem \
-out certificate.pem \
-days 365
Could not find private key from public_key.pem
000458984C7F0000:error:1608010C:STORE
routines:ossl_store_handle_load_result:unsupported:crypto/store/store_result.c:160:
provider=default
```

Usando a chave privada:

```
> openssl req -x509 \
-key private_key.pem \
-out certificate.pem \
-days 365
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BR
State or Province Name (full name) [Some-State]:SC
Locality Name (eg, city) []:Blumenau
Organization Name (eg, company) [Internet Widgits Pty Ltd]:MSA
Organizational Unit Name (eg, section) []:Monsters
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:msa@contact.com
```

Questão 4

Exiba os campos do certificado: versão, serial, emissor, validade, subject, extensões

```
> openssl x509 -in certificate.pem -text -noout
```

Certificate:

```
    Data:
      Version: 3 (0x2)
      Serial Number:
        4c:16:64:e2:61:09:c4:60:55:2a:43:7a:fe:b2:0f:8d:d0:85:63:6d
      Signature Algorithm: sha256WithRSAEncryption
      Issuer: C=BR, ST=SC, L=Blumenau, O=MSA, OU=Monsters, CN=localhost,
emailAddress=msa@contact.com
      Validity
        Not Before: Jun  1 23:26:32 2026 GMT
        Not After : Jun  1 23:26:32 2027 GMT
      Subject: C=BR, ST=SC, L=Blumenau, O=MSA, OU=Monsters, CN=localhost,
emailAddress=msa@contact.com
```

[...]

```
X509v3 extensions:
X509v3 Subject Key Identifier:
    8A:2A:8A:FB:36:FE:EA:7E:08:48:8F:86:5E:F6:76:EF:08:20:FA:5A
X509v3 Authority Key Identifier:
    8A:2A:8A:FB:36:FE:EA:7E:08:48:8F:86:5E:F6:76:EF:08:20:FA:5A
X509v3 Basic Constraints: critical
CA:TRUE
```

Questão 5

Converta o certificado para formato DER e compare o tamanho com o certificado no formato .PEM

certificate.der	1003B
certificate.pem	1.4K

.der: 1003 bytes

.pem: 1.4 KiB = 1415 bytes

Questão 6

Baixe o certificado “furb.pem” que está no AVA e exiba as propriedades do certificado. Identifique:

- Quem emitiu o certificado

C=BR, O=REDE NACIONAL DE ENSINO E PESQUISA - RNP, CN=RNP ICPEdu GR46 OV TLS CA 2025

- Período de validade

Not Before: May 12 11:11:15 2026 GMT

Not After : Nov 27 11:11:15 2026 GMT

- Nome alternativo do proprietário do certificado

DNS:*.furb.br, DNS:furb.br

Questão 7

A partir do certificado fornecido, identifique o campo “Authority Information Access”. Obtenha a cadeia de confiança (raiz > intermediário > usuário final)

Com o arquivo .pem conseguimos obter a informação:

RNP ICPEdu GR46 OV TLS CA 2025 > *.furb.br

Issuer: C=BR, O=REDE NACIONAL DE ENSINO E PESQUISA - RNP, CN=RNP ICPEdu GR46 OV TLS CA 2025

Authority Information Access:

CA Issuers -

URI:http://secure.globalsign.com/cacert/rnpicpedugr46ovtlsca2025.crt

OCSP - URI:http://ocsp.globalsign.com/rnpicpedugr46ovtlsca2025

Então conseguimos pegar o root com:

➤ wget http://secure.globalsign.com/cacert/rnpicpedugr46ovtlsca2025.crt

Converter para .pem com:

➤ openssl x509 -inform DER -in rnpicpedugr46ovtlsca2025.crt -out intermediate.pem

Ler o conteúdo:

```
› openssl x509 -in intermediate.pem -text -noout
```

```
Issuer: C=BE, O=GlobalSign nv-sa, CN=GlobalSign Root R46
```

```
Authority Information Access:
```

```
    OCSP - URI:http://ocsp.globalsign.com/rootr46
```

```
    CA Issuers - URI:http://secure.globalsign.com/cacert/rootr46.crt
```

Portanto a cadeia de confiança seria:

```
GlobalSign Root R46 > RNP ICPEdu GR46 OV TLS CA 2025 > *.furb.br
```